

## Introduction/Abstract/Problem Statement

### Motivation and Significance from Statement of Work:

The prevalence of diabetes, particularly Type-2 diabetes, pre-diabetes, and gestational diabetes, has seen a significant rise globally, affecting millions of individuals. Early detection and diagnosis of these conditions are crucial to managing and preventing serious complications such as cardiovascular diseases, neuropathy, and kidney failure. Despite advancements in medical diagnostics, many individuals remain undiagnosed or are diagnosed too late, leading to preventable health deterioration. This project aims to develop an AI-driven model that can predict the likelihood of an individual having pre-diabetes, Type-2 diabetes, or gestational diabetes using easily accessible user data, including demographics, lifestyle factors, and biometric measurements. It will also be able to give the appropriate guidance based on that prediction on how to improve the user's health. The impact of solving this problem could lead to earlier interventions, improved patient outcomes, and reduced healthcare costs, contributing significantly to public health.

### Implementation/Methodology

- Data collection/preprocessing

For our main survey data, we collected five different datasets from various sources such as Github, Kaggle, and the CDC. Some datasets had specific classes that were not found in others such as the PIMA Indians Dataset for gestational diabetes. The Behavioral Risk Factor Surveillance System (BRFSS) is a health-related telephone survey that is collected annually by the CDC. Each year, the survey collects responses from over 400,000 Americans on health-related risk behaviors, chronic health conditions, and the use of preventative services. It has been conducted every year since 1984. We got the 2021 version of the survey data and filtered out the features that were most relevant for diabetes diagnosis such as physical activity, diet, age, BMI, etc. After combining the data of all 5 datasets, we then separated our data based on gender; we had one dataset for male patients and one dataset for female patients. The reasoning behind this was because males don't get gestational diabetes, hence we had to train one structured data model for each gender. For those models, categorical features were label encoded using the Sklearn library in Python and numerical features were left as is. We also added a column 'Patient\_ID' to keep track of each patient's data while training and to organize our time series data.

- Data simulation for CGM data

For our Continuous Glucose Monitoring (CGM) data for aiding in our diagnoses, we completely simulated the data across all 4 classes of diagnoses: no diabetes, prediabetes, type 2 diabetes, and gestational diabetes. A group of 24 timestamps (represented 24 hours in a day) was generated for each male and female patient, with a corresponding glucose measurement for each timestamp. The 'Patient\_ID' column from the structured data of male and female patients were used to group the data by patient. For preprocessing, we converted the timestamps from time format (00:00:00) to total seconds after midnight and then scaled them using MinMax scaling which transformed each value between 0 and 1. We also scaled the glucose values the same way.

- Model selection
  - Male and Female – describe different models used and what you ended up with  
We tried several different models before settling on a final type. We tried XGBoost, Random Forest, Gradient Boosting, Linear SVM, and Stochastic Gradient Descent Logistic Regression. We trained the model with the structured data, preprocessed the same way as mentioned for each and for 100 epochs. We then graphed the accuracies and losses of each model after training. The results concluded that XGBoost and Random Forest were the best options, with XGBoost having a final accuracy of around 90% and Random Forest being suspiciously 100% accurate. To confirm, we did another test and compared XGBoost and Random Forest with each other. That result ended up siding with XGBoost, with an accuracy of 85%.
  - CGM model – describe different topologies and what you ended up with  
We experimented with different
- Workflow
  - Dataset/Database
  - Feature extraction
  - Preprocessing
  - Fitting models on data
  - Diabetes prediction

## Results

- Types of classification
- Recommendations
- Evaluation
  - Training/Validation Accuracy Curves for CGM, Male and Female models

- Classification Reports: Precision, Recall, F1-Score
- Testing of LLM
- Screenshot of app

## Conclusion (Impact and Future Work)

- Impact
  - Personalized approach to patient diabetes diagnosis
  - No cost
  - Eliminate need of going to labs for testing purposes
- Future Work
  - **\*Patient Compliance and Data Quality\***: Data quality and patient compliance are major challenges in healthcare. Patients may forget to log their meals or exercise activities, which can impact prediction accuracy. Future improvements should focus on integrating automatic data collection tools, such as wearable devices that track physical activity and dietary logs using smart cameras, to improve the quality and completeness of data.
  - **\*Scalability for Real-time Data Handling\***: A common problem in real-world applications is scaling the solution to accommodate multiple users at once, especially when dealing with real-time data. Future improvements should focus on developing a scalable backend infrastructure capable of handling simultaneous data streams from numerous users without compromising performance.
  - **\*User Engagement and Behavior Analysis\***: Use real-time data to engage users and analyze their behavior patterns. For example, tracking whether users follow recommendations and providing real-time feedback can help in maintaining better glycemic control. Future work should focus on creating a feedback loop that motivates users to adhere to healthy habits.
  - **\*Data Privacy and Security\***: Handling real-time health data comes with privacy and security concerns. Future improvements should involve the integration of encryption protocols and secure authentication mechanisms to ensure that users' real-time data is kept confidential and safe from breaches.

## Related Work (Literature Review)

## References

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[https://github.com/ThecoderPinar/Diabetes\\_Health\\_Prediction\\_and\\_Analysis](https://github.com/ThecoderPinar/Diabetes_Health_Prediction_and_Analysis)

cpatricksalves. (2019). GitHub - cpatricksalves/kaggle-diabetes-prediction: A Kaggle competition

promoted by Data Science Academy aimed to create a Machine Learning model to predict the occurrence of diabetes. GitHub. <https://github.com/cpatricksalves/kaggle-diabetes-prediction>

Video Plan:

## 1. Introduction/Abstract/Problem Statement (1 minute)

- **Prevalence of Diabetes:**
  - Significant rise in Type-2 diabetes, pre-diabetes, and gestational diabetes globally.
  - Millions of individuals affected, with serious complications like cardiovascular diseases, neuropathy, and kidney failure.
- **Problem:**
  - Late or missed diagnoses lead to preventable health deterioration.
- **Solution:**
  - AI-driven model to predict diabetes conditions (pre-diabetes, Type-2, and gestational) using accessible user data.
  - Aims to provide health improvement guidance based on predictions.
- **Impact:**
  - Early intervention, improved patient outcomes, reduced healthcare costs.

## 2. Motivation and Significance (1 minute)

- **Importance of Early Diagnosis:**
  - Early detection crucial for managing diabetes and avoiding complications.
  - Goal to make diagnostics accessible without the need for lab visits.
- **Public Health Contribution:**
  - Aimed at reducing the burden on healthcare systems and improving overall public health.

### 3. Implementation/Methodology (2 minutes)

- **Data Collection/Preprocessing:**
  - Collected datasets from multiple sources (GitHub, Kaggle, CDC).
  - Data filtering: focusing on features like physical activity, diet, age, BMI.
  - Separate datasets for male and female due to the gestational diabetes variable.
  - Categorical features label encoded; numerical features scaled using MinMaxScaler.
  - Added 'Patient\_ID' for tracking and organizing time series data.
- **Data Simulation for CGM Data:**
  - Simulated Continuous Glucose Monitoring (CGM) data for four diagnosis classes.
  - Generated 24 timestamps (24 hours) for each patient with corresponding glucose measurements.
  - Preprocessed and scaled timestamps and glucose data using MinMaxScaler.
- **Model Selection:**
  - Experimented with XGBoost, Random Forest, Gradient Boosting, Linear SVM, and SGD Logistic Regression.
  - Trained models on structured data for 100 epochs.
  - XGBoost performed best with around 90% accuracy, Random Forest gave suspiciously 100% accuracy (re-validated with XGBoost confirming better performance).

### 4. Workflow (30 seconds)

- **Data Flow:**
  - Datasets stored in a database.
  - Features extracted and preprocessed for model training.
  - Models fitted to data for diabetes prediction.

### 5. Results (30 seconds)

- **Types of Classification:**
  - Four classes: no diabetes, pre-diabetes, type 2 diabetes, gestational diabetes.
- **Model Performance Evaluation:**

- Training/validation accuracy curves for CGM, Male, and Female models.
- Classification reports including Precision, Recall, F1-Score for each model.
- Testing of LLM (Large Language Model) for additional insights.

## **6. Conclusion: Impact and Future Work (1 minute)**

- **Impact:**

- Personalized, no-cost approach to diabetes diagnosis.
- Eliminates the need for lab visits.
- Explain results of UI and Llama recc

- **Future Work:**

- **Patient Compliance and Data Quality:**
  - Integration of automatic data collection tools (wearables, smart cameras).
- **Scalability for Real-Time Data Handling:**
  - Improve backend infrastructure to handle simultaneous user data streams.
- **User Engagement and Behavior Analysis:**
  - Real-time feedback loop for better glycemic control.
- **Data Privacy and Security:**
  - Implement encryption and secure authentication for user data protection.